

Zeyad El-Sayed

AI Systems Engineer / LLM & Agentic AI

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Cairo, Egypt | Military Service: Completed

SUMMARY

AI Systems Engineer building production Machine Learning Systems for LLM and agentic infrastructure. Shipped an autonomous multi-agent research system on LangGraph with human-in-the-loop control and automatic 4-provider LLM failover, plus a multilingual RAG platform handling Arabic/English hybrid retrieval at 1–3s latency. Combines applied machine learning (PEFT fine-tuning, evaluation-driven prompt engineering) with async FastAPI, PostgreSQL/pgvector, and Redis backends deployed via Docker with CI/CD. NLP models contributed to **John Snow Labs’ Spark NLP** Hub.

SKILLS

Languages: Python, C++, SQL

LLM & Agentic AI: LangChain, LangGraph, LangSmith, CrewAI, RAG, Hybrid Retrieval, Query Rewriting, Cross-Encoder Reranking, Tool Calling, Prompt Engineering, PEFT (LoRA/QLoRA)

ML Frameworks: PyTorch, HuggingFace Transformers, Scikit-learn, Keras, NumPy, Pandas

Backend & APIs: FastAPI, Async APIs (AsyncSession, DI), Pydantic, REST, PostgreSQL, Redis

Infra & DevOps: Docker, Docker Compose, GitHub Actions (CI/CD), Linux, Railway, Cloudflare Workers

Vector & Retrieval: pgvector, BM25, RRF, Semantic Search

PROJECTS

TesseractResearch | *LangGraph, FastAPI, AsyncPostgreSQL (pgvector), Redis, Docker*

GitHub | Live Demo

- Architected a 6-stage autonomous research agent on **LangGraph** (planner → search → HITL → scraper → analyzer → report) with a self-correction evaluator that re-routes on failed quality checks.
- Built a 4-provider LLM failover dispatcher (Groq, Mistral, Cerebras, HuggingFace) with Redis-tracked credit zones to prevent quota exhaustion mid-run.
- Integrated 5 external tools for search and scraping (Tavily, Serper, Firecrawl, Jina, BeautifulSoup) into the agent’s tool-calling layer.
- Implemented human-in-the-loop approval via LangGraph’s `interrupt()` with **AsyncPostgresSaver** checkpointing for durable, resumable agent state.
- Shipped an async FastAPI backend (**AsyncSession** DI) in a 4-service Docker Compose stack (app, pgvector, Redis, RedisInsight) with GitHub Actions CI/CD.

TesseractRAG | *FastAPI, PostgreSQL, pgvector, BM25, Redis, Docker*

GitHub | Live Demo

- Redesigned v1’s flat architecture into a dispatcher/abstract-base-class pattern – any component (embedder, retriever, generator, DB) is swappable by adding one class.
- Built hybrid BM25 + pgvector retrieval fused via RRF for multilingual (Arabic/English) queries, at 1–3s response latency with automatic semantic-only fallback.
- Designed a 3-zone LLM routing layer tracking daily Groq quota in Redis (Groq-only → round-robin → HuggingFace fallback) with retrieval-specific query rewriting.
- Migrated from JSON + serialized FAISS to PostgreSQL + pgvector, eliminating partial-write corruption risk and adding ACID guarantees with Alembic migrations.
- Delivered a role-based analytics layer – admin dashboard and per-user stats panel – via JWT-scoped FastAPI endpoints.

Catch the AI (Graduation Project) | *Python, Django, React, PostgreSQL, ViT, DeBERTa, Wav2Vec 2.0*

GitHub | Website

- Trained ViT/EfficientNet models detecting AI-generated images at **96%** accuracy for media verification.
- Built an NLP-based text detector (DeBERTa, RoBERTa) reaching **95%** accuracy on AI-generated text.
- Applied Wav2Vec 2.0 for AI-generated audio detection at **90%** accuracy.

Fine-Tuning LLMs | *PyTorch, PEFT, HuggingFace Transformers, NLP*

GitHub | HF Spaces

- Curated a fine-tuning repository, deploying models on HF Spaces for classification, generation, and summarization.
- Contributed text-detection models to **John Snow Labs’ Spark NLP** Models Hub for distributed NLP use.
- Applied LoRA/QLoRA PEFT, cutting trainable parameters by up to 99% while maintaining 95%+ performance (ROUGE-L, BLEU).

LLMs from Scratch | *PyTorch, NLP, Transformers*

GitHub

- Implemented an LLM from scratch in pure PyTorch – tokenization, multi-head self-attention, positional encodings, full transformer blocks.
- Automated end-to-end training/evaluation pipelines to validate architecture internals.

EXPERIENCE

Information Technology Institute (ITI) Business Intelligence Trainee

Jul 2023 – Sep 2023 *Egypt*

- Built interactive Tableau/Power BI dashboards translating raw datasets into business insights.
- Applied ETL pipelines, dimensional modeling, and SQL-based data warehouse design.

EDUCATION

Benha University B.S. in Computer Science & Artificial Intelligence

Oct 2020 – Jun 2024 *GPA: 3.4 / 4.0*

LANGUAGES

Arabic: Native | **English:** Professional working proficiency